

# Cross-Domain Evaluation of DAViT

Testing Generalisation of a Pretrained Vision Transformer for Pneumonia Detection

**Haseeb Arshad (23i-2578) • Haider Farooq (23i-2542) • Mudassir Asghar (23i-2577)**

Department of AI & Data Science, FAST-NUCES Islamabad

# The Domain Gap Challenge

---

## Why it Matters?

- **Clinical Need:** Pneumonia kills 740,000+ annually; AI offers scalable diagnosis.
- **The Barrier:** Models trained at one hospital (Source) fail at another (Target).
- **DAViT's Claim:** Use DINOv2 backbone for "domain robustness."
- **Our Mission:** Conduct a Zero-Shot Transfer to test if it survives unseen data.

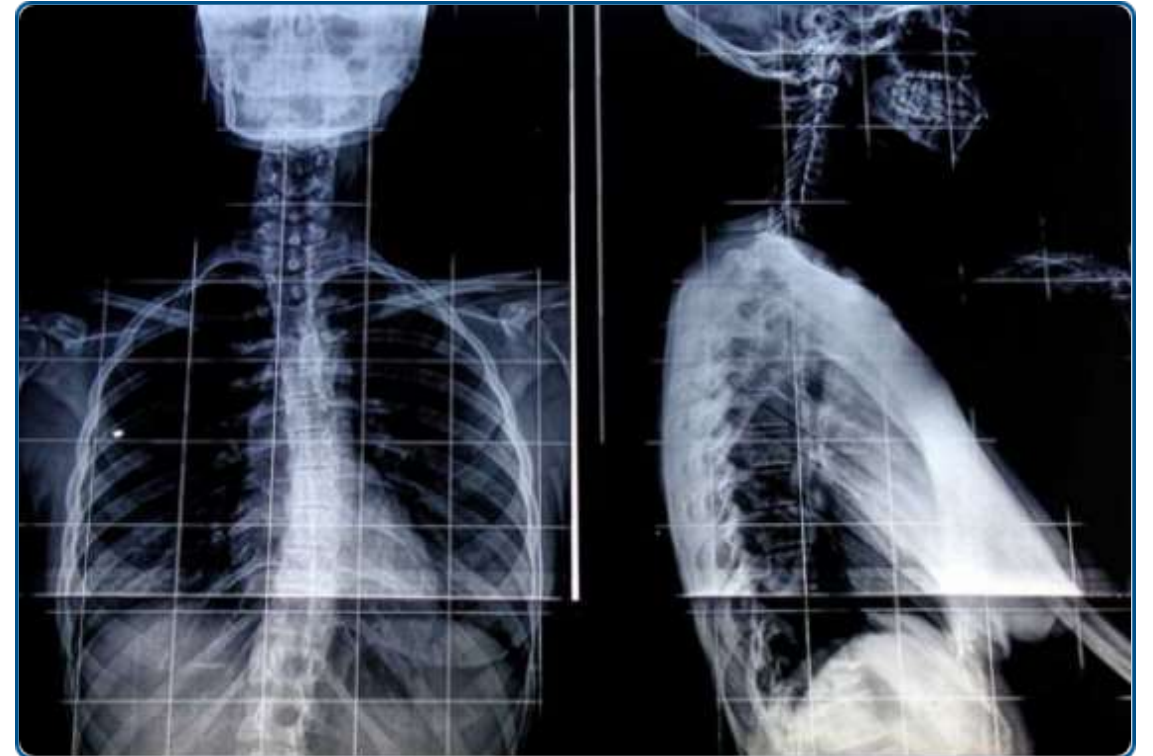


Fig 1: Pediatric (Kaggle) vs Adult (CoronaHack) Domain Differences

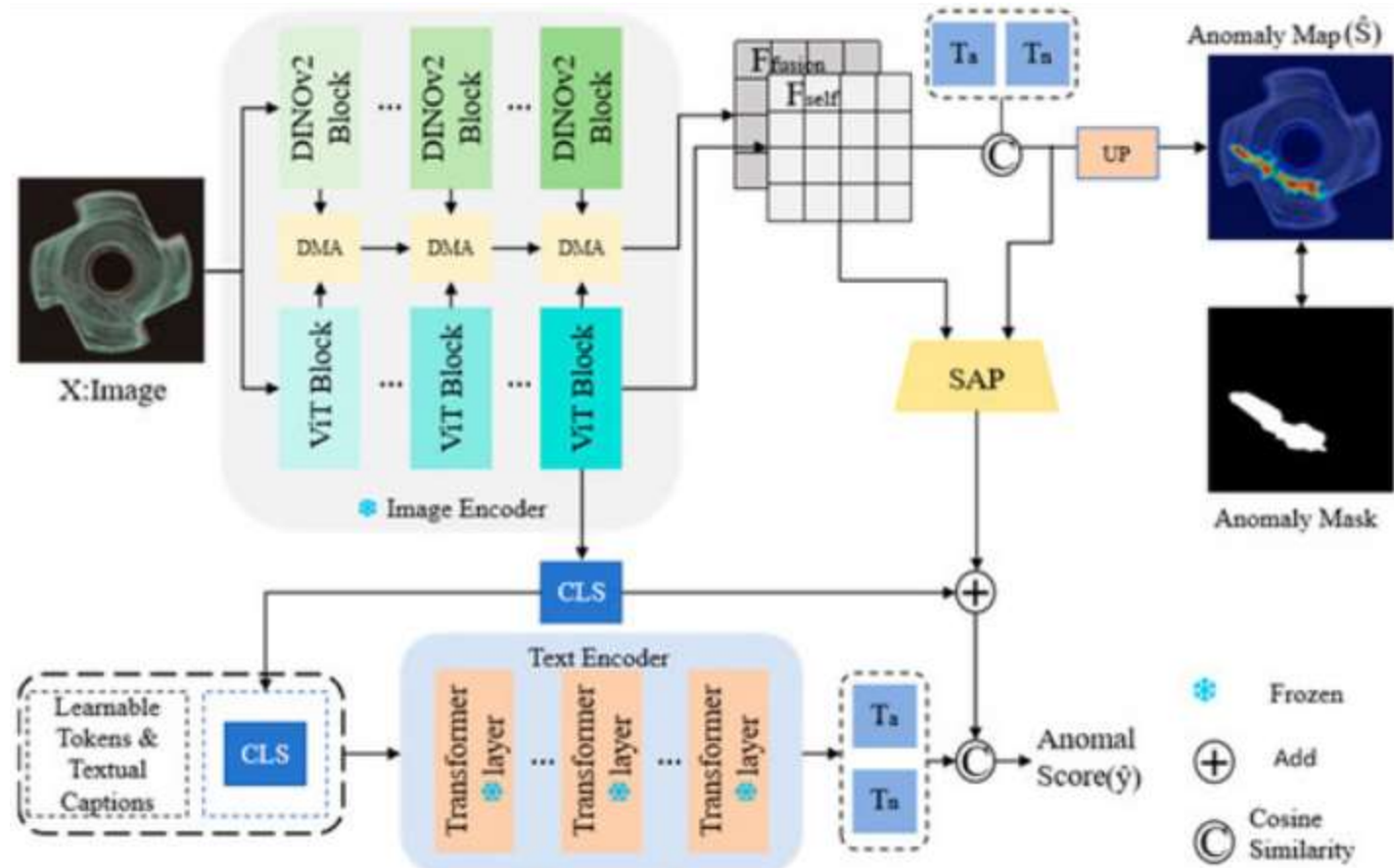
# Literature Review: Contemporary Methods

---

| Baseline Papers (2024–25)    | Technique                 | Limitations / Research Gaps                         |
|------------------------------|---------------------------|-----------------------------------------------------|
| DAVIT (Fu et al., 2025)      | DINOv2 + Domain Aug       | Only evaluated on in-distribution test sets.        |
| Inception-LSTM (Ghuse, 2025) | Spatial-Sequential Hybrid | Lacks Explainability; restricted to pediatric only. |
| VGG19 (Ennab, 2025)          | Grad-CAM vs Pixel-Level   | Dated architecture; high False Positive Rates.      |

Our Study: First Cross-Dataset Evaluation on CoronaHack for DAVIT.

# Proposed DANN + GRL Architecture



Objective:

$$L_{\text{total}} = L_{\text{class}} - \lambda \cdot L_{\text{domain}}$$

# The DAViT Pipeline & Failure Point



## Frozen Backbone

**DINOv2-Large:** 300M params. Frozen to preserve self-supervised general features.

**Input:** CXR 224x224

**Output:** 1024-dim [CLS] Token

## Failure: The CNN Head

A 3-layer FC head trained on Kaggle. It memorized the **texture** of pediatric scanners.

**Result:** Miscalibration when seeing unseen adult imaging domains.

# Three-Way Comparison (F1-Score)

---



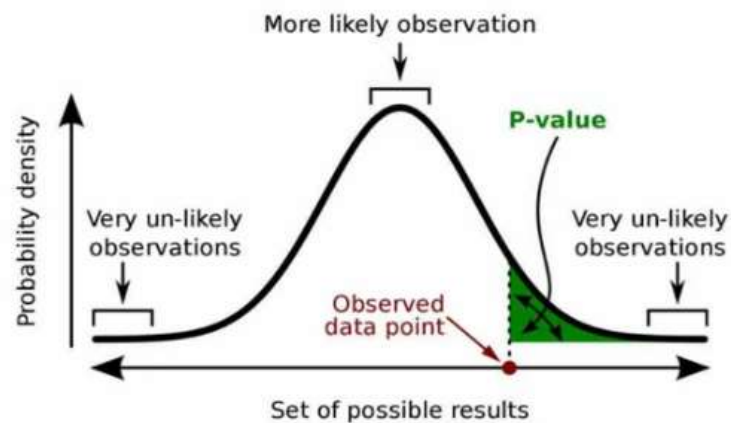
**-14.9% Drop**

F1 Degradation under Domain Shift

**10x FP Spike**

Increased False Positives on Unseen Data

# Statistical Significance Analysis



A **p-value** (shaded green area) is the probability of an observed (or more extreme) result assuming that the null hypothesis is true.

## Two-Proportion Z-Test

We tested the null hypothesis: "Recall rates are equal across domains."

- **Z-Score:** 2.87
- **Result:** Reject  $H_0$  at  $\alpha = 0.01$

**Conclusion:** Degradation is statistically real, not random sampling noise.

# Why Does it Fail? (Discussion)

---



## Texture Overfit

CNN Head memorized pediatric scanner calibration. Adult scanners differ in contrast.



## Inverted Imbalance

Kaggle ratio (0.6:1) vs CoronaHack (1.8:1). Boundary biased toward viral class.



## Heterogeneity

Multi-source hospital systems introduce acquisition noise DINOv2 can't fully fix.

# Key Contributions & Outcomes

---

## Evaluation

First Cross-Dataset evaluation of DAViT on an entirely unseen Adult domain.

## Failure Analysis

Identified CNN Head as the primary failure point for Vision Transformers in Med-AI.

## Domain Shift

Statistically confirmed domain shift with Z-test ( $p=0.0041$ ).

## Architecture

Proposed DANN + GRL architecture as the remedy for hospital-agnostic deployment.

# Questions?

**Thank you for your time and attention**

Haseeb Arshad – 23i-2578

Haider Farooq – 23i-2542

Muddassir Asghar – 23i-2577